🌾 AgroBrain 2.0

AI-Enhanced Smart Farming Advisory System

🌍 Location-Based Weather Intelligence

Enter City Name

chennai

Weather Forecast Analyzed for chennai

🌡️ 5-Day Avg Temperature: 26.48 °C

💧 5-Day Avg Humidity: 68.55 %

🌧️ Avg Rainfall per Day (5-Day): 0.0 mm

📅 Estimated 30-Day Rainfall Projection: 0.0 mm

🌱 Soil Health Card Inputs

Nitrogen Level (SHC)

Low

Phosphorus Level (SHC)

High

Potassium Level (SHC)

High

Soil pH

7.01

Soil Condition

Moist

Analyze Farm Conditions

🌱 Top 5 Crop Recommendations

1. **Grapes** — 26.50% suitability
2. **Muskmelon** — 23.00% suitability
3. **Lentil** — 16.50% suitability
4. **Chickpea** — 13.00% suitability
5. **Mothbeans** — 6.50% suitability

📋 Advisory for Top 2 Crops

🌾 Grapes

🦠 Disease Risk: **High**

💧 Irrigation Advice: **2-3 days**

🌾 Muskmelon

🦠 Disease Risk: **High**

💧 Irrigation Advice: **3-4 days**

🤖 AI Explanation

Why Grapes is Suitable 🍇

Grapes are a fantastic choice for this farm due to the warm and dry weather. With an average temperature of 26.48 °C and low rainfall, grapes will thrive in these conditions. Additionally, the soil pH of 7.01 is ideal for grape cultivation, as it's close to the optimal range of 6.5-7.5. The soil's moisture level and nutrient availability (Nitrogen: 30, Phosphorus: 100, Potassium: 150) will also support healthy grape growth.

Disease Prevention Measures for High Risk Level 🚨

To prevent diseases, implement the following high-risk measures:

- Regularly inspect the vines for signs of fungal infection or pests.
- Use integrated pest management (IPM) techniques to control any infestations.
- Apply fungicides and bactericides as needed to prevent disease spread.
- Maintain good air circulation around the vines to reduce humidity and prevent fungal growth.

Irrigation Advice for 3-4 Days 💧

Given the current dry spell, it's essential to irrigate the grapevines every 3-4 days to maintain soil moisture. Use drip irrigation or sprinklers to deliver water directly to the roots, reducing evaporation and runoff. Aim to provide about 20-30 mm of water per irrigation cycle, depending on soil type and temperature.

Rainfall Caution Considering 30-Day Rainfall of 0.0 mm ☁️

With no rainfall in the past 30 days, be cautious of soil compaction and water stress. Monitor soil moisture levels closely and adjust irrigation schedules accordingly. Also, be prepared for potential heat stress in the grapevines, which can lead to sunburn and reduced yields.